

Quiz CF1Q

Question 1

Incorrect

Mark 0.00 out of 2.00

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Let $f: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^3$ and $g: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \lceil x \rceil$.

Is the following argument logically correct?

The functions f and g are both continuous at $x = 6$. Therefore, also $f - g$ is continuous at $x = 6$.

☐ No

☒ Yes

✗

The correct answer is: No

Die Antwort ist falsch.

Question 2

Incorrect

Mark 0.00 out of 2.00

[Flag question](#)

Consider the functions defined on $\mathbb{R}$:

$$a(x) = H(x - 3) + 4$$

$$b(x) = \begin{cases} 13(x - 3) & \text{if } x < 3 \\ \sin(x - 3) - 8 & \text{if } x \geq 3 \end{cases}$$

$$c(x) = e^{x-3}$$

$$d(x) = \begin{cases} 13(x - 3) - 4 & \text{if } x < 3 \\ \cos(x - 3) + 3 & \text{if } x \geq 3 \end{cases}$$

$$e(x) = \lfloor x - 3 \rfloor$$

Choose two of the given functions such that

- both are **discontinuous** at $x = 3$ and
- their **difference** is **continuous** at $x = 3$.

☒ c ✗

☐ a

☐ b

☒ e ✓

☐ Two such functions cannot exist.

☐ $f(x) := 1.62$

☐ d

Die Antwort ist falsch.

The correct answers are: a, e

Question 3

Correct

Mark 2.00 out of 2.00

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Let $f: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \begin{cases} 1(x + 1) + 6 & \text{if } x < -1 \\ 10.5 & \text{if } -1 \leq x \leq 9 \\ (x - 9)^7 + 15 & \text{if } x > 9 \end{cases}$.

Determine the largest set U , such that f is continuous on U .

☐ $\mathbb{R}$

☐ $\mathbb{R} \setminus \{9\}$

☐ $\mathbb{R} \setminus \{-1\}$

☒ $\mathbb{R} \setminus \{-1, 9\}$

✓

The correct answer is: $\mathbb{R} \setminus \{-1, 9\}$

Die Antwort ist richtig.

Question 4

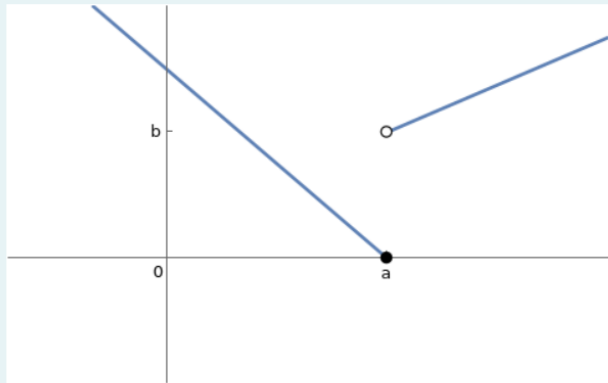
Partially correct

Mark 2.00 out of 4.00

Flag question

For $a, b \in \mathbb{R}$ consider the function

$$G_{a,b}(x) := \begin{cases} |7(x-a)| & \text{if } x \leq a \\ |3(x-a)| + b & \text{if } x > a \end{cases}$$



Now take $a = 2$ and $b = 9$. From the graphical representation of the function one may conclude that $G_{2,9}$ is at $x_0 =$

.

In order to **prove this using the ϵ - δ -criterion**, we have to show that

$$\exists \epsilon > 0 \forall \delta > 0 \exists x \in \mathbb{R} : |x - 2| < \delta \wedge |G_{2,9}(x) - G_{2,9}(2)| \geq \epsilon.$$

This means, we have to **first find an $\epsilon > 0$** and **then find** (for an arbitrary $\delta > 0$) an $x \in \mathbb{R}$ such that

$$|x - 2| < \delta \wedge |G_{2,9}(x)| \geq \epsilon.$$



We show that the function is discontinuous at $x_0 = 2$.

The correct answer is: discontinuous, 2

1. Which of the following is a **suitable choice for ϵ** ?

- ☐ $\epsilon = 10$
- ☒ $\epsilon = 9$
- ☐ $\epsilon = 27$
- ☐ $\epsilon = \frac{x}{5\delta}$



The correct answer is: $\epsilon = 9$

2. Let now $\delta > 0$ arbitrary but fixed. Which of the following is a **suitable choice for x** that **proves** that $G_{2,9}$ is **discontinuous** at $x_0 = 2$?

- ☐ $x = 2 + \frac{1}{5}$
- ☐ $x = 2 + \frac{\delta}{5}$
- ☐ $x = \frac{\delta}{5}$
- ☒ $x = 2 - \frac{\delta}{5}$



The correct answer is: $x = 2 + \frac{\delta}{5}$

Question 1

Quiz CF1B

Partially correct

Mark 0.33 out of 1.00

For $a \in \mathbb{R}$ consider the *modified Heaviside function*

$$H_a(x) := \begin{cases} 1 & \text{if } x > a \\ 0 & \text{if } x \leq a \end{cases}$$

Now take $a = 66$. From the graphical representation of the function one may conclude that H_{66} is at x_0 if and only if x_0 .



The correct answer is: continuous, ≠ 66

1. Now we show that H_{66} is *discontinuous* at $x_0 = 66$ using the definition of continuity directly. Which of the following sequences proves the discontinuity of H_{66} at $x_0 = 66$?

- ☐ $x_n = 66 + \frac{7}{n}$
☐ $x_n = 132 + \frac{1}{n}$
☐ $x_n = 66 + e^{-n}$
☒ $x_n = 66 - \frac{1}{n^2}$



The correct answer is: $x_n = 66 + e^{-n}$

2. We show that H_{66} is *continuous* at $x_0 \neq 66$ by the ε - δ -criterion. Assume $x_0 \neq 66$ and $\varepsilon > 0$ arbitrary but fixed. We have to find some $\delta > 0$ such that, for all $x \in \mathbb{R}$, $|x - x_0| < \delta$ enforces

$$|H_{66}(x) - H_{66}(x_0)| < \varepsilon.$$

Which of the following is a suitable choice for δ ?

- ☐ $\delta = 2|x_0 - 66|$
☐ $\delta = |x_0 + 66|$
☐ $\delta = \frac{1}{6}|x_0 - 66|$
☒ $\delta = \frac{\varepsilon}{7}$



The correct answer is: $\delta = \frac{1}{6}|x_0 - 66|$

Die Antwort ist teilweise richtig.

You have correctly answered 1 part(s) of this question.

Question 2

Partially correct

Mark 0.67 out of 1.00

For $a, b \in \mathbb{R}$ consider the function

$$G_{a,b}(x) := \begin{cases} |9(x-a)| & \text{if } x \leq a \\ |2(x-a)| + b & \text{if } x > a \end{cases}$$

Now take $a = 9$ and $b = 1$. From the graphical representation of the function one may conclude that $G_{9,1}$ is continuous at $x_0 \neq 9$. In order to prove this using the ϵ - δ -criterion, we choose $x_0 \neq 9$ and $\epsilon > 0$ arbitrary but fixed and show

$$\exists \delta > 0 \forall x \in \mathbb{R} : |x - x_0| < \delta \implies |G_{9,1}(x) - G_{9,1}(x_0)| < \epsilon.$$

In order to find such $\delta > 0$, we distinguish whether x_0 is on the left or right branch of the function.

1. Assume $x_0 < 9$: For any $x \in \mathbb{R}$ we observe the following: if $|x - x_0| < |x_0 - 9|$ then $x < 9$ and therefore $G_{9,1}(x) =$, which implies

$$\begin{aligned} |G_{9,1}(x) - G_{9,1}(x_0)| &= ||9(x-9)| - |9(x_0-9)|| \\ &= 9 ||x-9| - |x_0-9|| \\ &= 9 ||x-x_0+x_0-9| - |x_0-9|| \\ &\leq 9(|x-x_0| + |x_0-9| - |x_0-9|) = 9|x-x_0|. \end{aligned}$$

✓

The correct answer is: |9(x - 9)|

2. Assume $x_0 > 9$: For any $x \in \mathbb{R}$ we observe the following: if $|x - x_0| < |x_0 - 9|$ then and therefore $G_{9,1}(x) = |2(x-9)| + 1$, which implies

$$\begin{aligned} |G_{9,1}(x) - G_{9,1}(x_0)| &= ||2(x-9)| + 1 - |2(x_0-9)| - 1| \\ &= 2 ||x-9| - |x_0-9|| \\ &= 2 ||x-x_0+x_0-9| - |x_0-9|| \\ &\leq 2(|x-x_0| + |x_0-9| - |x_0-9|) = 2|x-x_0|. \end{aligned}$$

✓

The correct answer is: x > 9

Which of the following is a suitable choice for $\delta > 0$, which allows to prove

$$\forall x \in \mathbb{R} : |x - x_0| < \delta \implies |G_{9,1}(x) - G_{9,1}(x_0)| < \epsilon,$$

i.e., the chosen δ must be such that, for an arbitrary but fixed $x \in \mathbb{R}$ with $|x - x_0| < \delta$, we have $|G_{9,1}(x) - G_{9,1}(x_0)| < \epsilon$.

- ☒ $\delta = \min(|x_0 - 9|, \frac{\epsilon}{2})$
- ☐ $\delta = \min(|x - 9|, \frac{\epsilon}{9}, \frac{\epsilon}{2})$
- ☐ $\delta = \max(|x_0 - 9|, \frac{\epsilon}{9}, \frac{\epsilon}{2})$
- ☐ $\delta = \min(|x_0 - 9|, \frac{\epsilon}{9}, \frac{\epsilon}{2})$

✗

The correct answer is: $\delta = \min(|x_0 - 9|, \frac{\epsilon}{9}, \frac{\epsilon}{2})$

Die Antwort ist teilweise richtig.

You have correctly answered 2 part(s) of this question.

Quiz CF2Q

Question 1

Incorrect

Mark 0.00 out of 3.00

[Flag question](#)

Let

$$f : \mathbb{R} \setminus \left\{ -3, -\frac{3}{5} \right\} \rightarrow \mathbb{R}, x \mapsto \frac{x+3}{-5x^2-18x-9}$$

Which of the following limits is $+\infty$?

- ☐ $\lim_{x \rightarrow -3/5} f(x)$
- ☐ $\lim_{x \rightarrow (-3/5)^-} f(x)$
- ☐ $\lim_{x \rightarrow -3} f(x)$
- ☐ $\lim_{x \rightarrow -3} f(x)$
- ☐ $\lim_{x \rightarrow -3^+} f(x)$
- ☒ $\lim_{x \rightarrow (-3/5)^+} f(x)$

✖

The correct answer is: $\lim_{x \rightarrow (-3/5)} f(x)$

Your answer is incorrect.

Question 2

Correct

Mark 3.00 out of 3.00

[Flag question](#)

Let

$$f : \mathbb{R} \setminus \left\{ \frac{5}{3} \right\} \rightarrow \mathbb{R}, x \mapsto \frac{3x^2+7x-20}{3x-5}$$

Compute $\lim_{x \rightarrow \frac{5}{3}} f(x) =$ (round to 2 decimal places and use "." or write number as a fraction with "/")

✓

The correct answer is: 5.67

Your answer is correct.

Question 3

Correct

Mark 2.00 out of 2.00

[Flag question](#)

Consider the following calculation:

$$\lim_{x \rightarrow 0} \operatorname{sgn} \left(\sin(x) + \left(\frac{6x^2 - 2x + 3}{x^6 + 36} \right) \right) \stackrel{(1)}{=} \operatorname{sgn} \left(\lim_{x \rightarrow 0} \sin(x) + \lim_{x \rightarrow 0} \left(\frac{6x^2 - 2x + 3}{x^6 + 36} \right) \right) =$$
$$\stackrel{(2)}{=} \operatorname{sgn} \left(0 + \frac{1}{12} \right) \stackrel{(3)}{=} \operatorname{sgn} \left(\frac{1}{12} \right) \stackrel{(4)}{=} 1$$

- ☐ Incorrect because in step 1 we must not push the "lim" inside the "sgn"-function
- ☐ Incorrect because in step 2 there is an error in the calculation of the limits
- ☒ Perfectly correct! ✓
- ☐ Incorrect because in step 3 there is an error in the calculation
- ☐ Incorrect because in step 4 there is an error in the calculation of the "sgn"-function
- ☐ Incorrect because in step 1 we must not split the "lim" into a sum of limits

Your answer is correct.

The correct answer is: Perfectly correct!

Question 4

Partially correct

Mark 0.67 out of 2.00

[Flag question](#)

Select all values that are accumulation points of $[-9, 6)$.

- ☐ -36
- ☐ -15
- ☐ -1
- ☐ -9
- ☐ 24
- ☒ 6 ✓
- ☐ 16

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: -9, 6, -1

Question 1

Correct

Mark 1.00 out of 1.00

Consider the functions

$$f_1 : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \begin{cases} 9e^x & \text{if } x < 0 \\ -7\sin(x) + 9 & \text{if } x \geq 0 \end{cases}$$

$$f_2 : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \begin{cases} 9\ln(x+1) & \text{if } x \leq 0 \\ -7x^9 + 4 & \text{if } x > 0 \end{cases}$$

Which of the functions f_i satisfies

$$\lim_{x \rightarrow 0^-} f_i = 9,$$

$$\lim_{x \rightarrow 0^+} f_i = 9, \text{ and}$$

$$\lim_{x \rightarrow 0} f_i = 0$$

- ☐ f_2
☐ f_1
☐ None of them, because the *two one-sided limits* cannot be equal.
☒ None of them, because if the *limit* is 0 the two *one-sided limits* cannot be equal to 9.



The correct answer is: None of them, because if the *limit* is 0 the two *one-sided limits* cannot be equal to 9.

Your answer is correct.

Question 2

Correct

Mark 1.00 out of 1.00

Which of the following sets has $\mathbb{R} \cup \{-\infty, \infty\}$ as its set of accumulation points?

- ☐ $\mathbb{R} \setminus (-6, 3]$
☐ $\mathbb{R}^+$
☐ $\{-6, 3\}$
☒ $\mathbb{R} \setminus \{-6, 3, 12\}$



The correct answer is: $\mathbb{R} \setminus \{-6, 3, 12\}$

Your answer is correct.

Quiz CF3Q

Question 1

Incorrect

Mark 0.00 out of 2.00

[Flag question](#)

Consider the functions

$$g_1 : [-9, 9] \rightarrow \mathbb{R}, x \mapsto \begin{cases} 9e^x & \text{if } x < 0 \\ 7x^9 + 9 & \text{if } x \geq 0 \end{cases}$$

$$g_2 : [-11, 11] \rightarrow \mathbb{R}, x \mapsto \begin{cases} 9 \ln(x+1) - 7 & \text{if } x \leq 0 \\ 7 \sin(x) - 7 \cos(x) & \text{if } x > 0 \end{cases}$$

Which of the functions g_i is *uniformly continuous* but *not continuous*?

- ☐ g_1
- ☒ g_2
- ☐ None of them, because every uniformly continuous function would also be continuous.
- ☐ None of them, because both are not uniformly continuous.

✖

The correct answer is: None of them, because every uniformly continuous function would also be continuous.

Your answer is incorrect.

Question 2

Incorrect

Mark 0.00 out of 2.00

[Flag question](#)

Consider the function $f : [-1, 1] \rightarrow \mathbb{R}, x \mapsto \sin(\pi(x + 0.74))^2 + 8.7$. Find the biggest interval $I \subseteq [-1, -0.11]$, such that f is **strictly increasing** on I .

$I = [$ $,$ $]$

✖

The correct answer is: -0.74, -0.24

Your answer is incorrect.

Question 2

Incorrect

Mark 0.00 out of 2.00

[Flag question](#)

Consider the function $f : [-1, 1] \rightarrow \mathbb{R}, x \mapsto \sin(\pi(x + 0.74))^2 + 8.7$. Find the biggest interval $I \subseteq [-1, -0.11]$, such that f is **strictly increasing** on I .

$I = [$ $,$ $]$

✖

The correct answer is: -0.74, -0.24

Your answer is incorrect.

Question 3

Correct

Mark 3.00 out of 3.00

[Flag question](#)

Consider the function

$$f : (0, 2] \rightarrow \mathbb{R}, x \mapsto \frac{5}{x} + 4.$$

We show that f is **not uniformly continuous** :

1. In order to prove that f is not uniformly continuous, we need to find two sequences (x_n) and (y_n) in $(0, 2]$, s.t.

$$\lim_{n \rightarrow \infty} |x_n - y_n| = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} |f(x_n) - f(y_n)| \neq 0.$$

2. Let now $x_n := \frac{1}{n}$ and $y_n := \frac{2}{n}$.
3. It is straightforward to see that

$$\lim_{n \rightarrow \infty} |x_n - y_n| = \lim_{n \rightarrow \infty} \left| \frac{1}{n} - \frac{2}{n} \right| = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

4. We also see that

$$\begin{aligned} \lim_{n \rightarrow \infty} |f(x_n) - f(y_n)| &= \lim_{n \rightarrow \infty} \left| 5n + 4 - \frac{5n}{2} - 4 \right| = \\ &= \lim_{n \rightarrow \infty} \frac{5n}{2} = \infty \neq 0. \end{aligned}$$

5. This shows that f is not uniformly continuous!

On the other hand, a theorem in the lecture notes says that **every continuous function on a closed interval must also be uniformly continuous**. Since this applies to f , there must be an error in the "proof" given above. What is the error?

- ☐ In **step 1**, the formula does not correspond to what it means to be not uniformly continuous.
- ☐ In **step 2**, the sequences do not have values in $(0, 2]$.
- ☐ In **step 3**, the limit calculation is incorrect.
- ☐ In **step 4**, the limit calculation is incorrect.
- ☒ The proof is correct, but the **theorem mentioned does not apply** to this example. ✓

Your answer is correct.

The correct answer is: The proof is correct, but the **theorem mentioned does not apply** to this example.

Question 4

Incorrect

Mark 0.00 out of 3.00

🚩 Flag question

Let $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto 2|x - 1| - 1|x| - 3|x + 1| + 7$. Note that $f(-4) = 4$ and $f(4) = -6$. Note also that the function is linear in any interval bounded by consecutive integers between -4 and 4. Compute a number w such that $f(w) = 1.5$.

$w =$

✗

Substitute your value w for x in the given function f , and you will see it does not give 1.5.

Please neglect the "correct answer" given below, we cannot show the correct answer here.

One possible correct answer is: 42

Your answer is incorrect.

Question 1

Correct

Mark 2.00 out of 2.00

Consider the function

$$f : [0, 9] \rightarrow \mathbb{R}, x \mapsto 5 \ln(x) + 5.$$

We show that f is **not uniformly continuous** in a similar way as it is done for $f(x) = x^2$ in the lecture notes.

1. In order to prove that f is not uniformly continuous, we need to find two sequences (x_n) and (y_n) in $[0, 9]$, s.t.

$$\lim_{n \rightarrow \infty} |x_n - y_n| = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} |f(x_n) - f(y_n)| \neq 0.$$

2. Let now $x_n := \frac{1}{n}$ and $y_n := \frac{2}{n}$.

3. It is straightforward to see that

$$\lim_{n \rightarrow \infty} |x_n - y_n| = \lim_{n \rightarrow \infty} \left| \frac{1}{n} - \frac{2}{n} \right| = \lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

4. We also see that

$$\begin{aligned} \lim_{n \rightarrow \infty} |f(x_n) - f(y_n)| &= \lim_{n \rightarrow \infty} |5 \ln(1/n) + 5 - 5 \ln(2/n) - 5| = \\ &= \lim_{n \rightarrow \infty} 5 \ln\left(\frac{1/n}{2/n}\right) = 5 \ln(1/2) \neq 0. \end{aligned}$$

5. This shows that f is not uniformly continuous!

On the other hand, a theorem in the lecture notes says that **every continuous function on a closed interval must also be uniformly continuous**. Since this applies to f , there must be an error in the "proof" given above. What is the error?

- ☐ In **step 1**, the formula does not correspond to what it means to be not uniformly continuous.
- ☐ In **step 2**, the sequences are not valid to serve as counter-examples because they must be sequences in $[0, 9]$.
- ☐ In **step 2**, the sequences are not valid to serve as counter-examples because we must take sequences that converge to ∞ .
- ☐ In **step 3**, the limit calculation is incorrect.
- ☐ In **step 4**, the limit calculation is incorrect.
- ☒ The **proof is correct**, but the **theorem mentioned does not apply** because the logarithm function cannot be defined on the closed interval $[0, 9]$. ✓

Your answer is correct.

The correct answer is: The **proof is correct**, but the **theorem mentioned does not apply** because the logarithm function cannot be defined on the closed interval $[0, 9]$.

Quiz DC1Q

Question 1

Incorrect

Mark 0.00 out of 2.50

🚩 [Flag question](#)

The graph of which function describes the **tangent line** of $f(x) := \ln x$ at 9

- ☐ $T(x) = \ln 9 + x$
- ☐ $T(x) = \ln 9 + 1 - \frac{x}{9}$
- ☐ $T(x) = \ln 9 - 1 + \frac{x}{9}$
- ☒ $T(x) = \frac{\ln 9}{9} + x$

✖

The correct answer is: $T(x) = \ln 9 - 1 + \frac{x}{9}$

Die Antwort ist falsch.

Question 2

Incorrect

Mark 0.00 out of 2.50

🚩 [Flag question](#)

Given the function $f(x) := x^2 + 2x + 72$, we compute its derivative at 8:

$$f'(8) \stackrel{(1)}{=} \lim_{h \rightarrow 0} \frac{(8+h)^2 + 2(8+h) + 72 - 64 - 16 - 72}{h}$$

$$\stackrel{(2)}{=} \lim_{h \rightarrow 0} \frac{h^2 + 17h + 2h}{h}$$

$$\stackrel{(3)}{=} \lim_{h \rightarrow 0} (h + 17 + 2)$$

$$\stackrel{(4)}{=} 19$$

Which of the equalities is incorrect

- ☐ (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☒ Computation is correct

✖

The correct answer is: (2)

Die Antwort ist falsch.

Question 3

Correct

Mark 2.50 out of 2.50

🚩 [Flag question](#)

Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ be the function

$$f(x) = \begin{cases} x^2 - 3x + 30 & \text{if } x < 3 \\ 3x + 21 & \text{if } x \geq 3 \end{cases}$$

Which of the following statements is **correct**?

- ☐ The function is **differentiable** but **not continuous** at 3.
- ☐ The function is **neither differentiable nor continuous** at 3.
- ☐ The function is **not differentiable** but **continuous** at 3.
- ☒ The function is **differentiable** and **continuous** at 3. ✓

Your answer is correct.

The correct answer is: The function is **differentiable** and **continuous** at 3.

Question 4

Incorrect

Mark 0.00 out of
2.50[Flag question](#)

The function $f(x) := x^3 - 48x - 32$ is strictly increasing at

- ☐ 3.65
- ☒ 4
- ☐ -4.11
- ☐ -3.62

✖

The correct answer is: -4.11

Die Antwort ist falsch.

Question 1

Correct

Mark 1.00 out of 1.00

Let $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ be the function

$$f(x) = \begin{cases} x^2 - 5x + 103 & \text{if } x < 10 \\ 15x + 3 & \text{if } x \geq 10 \end{cases}$$

Which of the following statements is **correct**?

- ☒ The function is **differentiable** and **continuous** at 10. ✓
- ☐ The function is **not differentiable** but **continuous** at 10.
- ☐ The function is **neither differentiable nor continuous** at 10.
- ☐ The function is **differentiable** but **not continuous** at 10.

Your answer is correct.

The correct answer is: The function is **differentiable** and **continuous** at 10.

Question 2

Correct

Mark 1.00 out of 1.00

Consider the function $f : [-2, 3] \rightarrow \mathbb{R}$, whose graph looks as follows:

- ☒ $f'(a) < f'(b)$ for all $a \in (-1, 2)$ and $b \in (2, 3)$
- ☐ $f'(-1.2) < 0$
- ☐ $f'(2.35) < 0$
- ☐ $f'(a) > f'(b)$ for some $a \in (-1, 2)$ and $b \in (2, 3)$
- ☐ $f'(-1) = f'(0)$

✓

The correct answer is: $f'(a) < f'(b)$ for all $a \in (-1, 2)$ and $b \in (2, 3)$

Your answer is correct.

Quiz DC2Q

Question 1

Correct

Mark 2.00 out of 2.00

🚩 Flag question

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \frac{4}{\pi + 16} x^2 \sin(5x - 8) + 17.$$

Calculate the derivative of f at $x = \frac{\pi+16}{10}$.

$$f' \left(\frac{\pi+16}{10} \right) = \boxed{4/5}$$

Write the answer as a **decimal number** or a **formula** involving "sin", "cos", and "pi".

✓

The correct answer is: 0.8

Your answer is correct.

Question 2

Correct

Mark 3.00 out of 3.00

🚩 Flag question

Given $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \cos(4x)$.

Compute $f^{(10)}(x)$.

$$f^{(10)}(x) = \boxed{-4^{10}} \cdot \cos(\boxed{4x})$$

✓

The correct answer is: (-1)^5*4^10, 4*x

(Note that you can enter big numbers as formulas like "362*2789" or "4^35".)

Your answer is correct.

Question 3

Correct

Mark 3.00 out of 3.00

🚩 Flag question

The function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - 2x^2 + 7$ has stationary points.

Let u be the largest number at which f is stationary. Compute u .

$$u = \boxed{4/3}$$

✓

The correct answer is: 1.3333333333333

The stationary point is ...

- ☒ a local minimum.
- ☐ a local maximum.
- ☐ neither local minimum nor local maximum.

✓

The correct answer is: a local minimum.

Your answer is correct.

Question 4

Partially correct

Mark 1.33 out of 2.00

[Flag question](#)

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions at $x \in \mathbb{R}$. Select the correct statement(s) from the following options:

- ☐ Suppose that f is differentiable at $g(x) \in \mathbb{R}$. Then

$$(f \circ g)'(x) = g'(f(x)) \cdot f'(x).$$

- ☒ Let $r \in \mathbb{R}$. Then ✗

$$f'(r \cdot x) = r \cdot f'(x).$$

- ☐ It is true that

$$(f \cdot g)'(x) = f'(x) \cdot g(x) - f(x) \cdot g'(x).$$

- ☒ It is true that ✓

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g(x)^2}.$$

- ☒ It is true that ✓

$$(f + g)'(x) = f'(x) + g'(x).$$

Your answer is partially correct.

You have selected too many options.

The correct answers are: It is true that

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g(x)^2}.$$

, It is true that

$$(f + g)'(x) = f'(x) + g'(x).$$

Question 1

Correct

Mark 1.00 out of 1.00

Given the function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^3 + 15x^2 - 168x + 19$. Calculate the **local minimum** and the **local maximum** of f .

The **local minimum** is at $x =$.



The correct answer is: 4

The **local maximum** is at $x =$.



The correct answer is: -14

Your answer is correct.

Question 2

Correct

Mark 1.00 out of 1.00

Given $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = e^{5x}$.

Compute $f^{(12)}(x)$.

$f^{(12)}(x) =$ $\cdot e^{$.



The correct answer is: 5^12, 5*x

(Note that you can enter big numbers as formulas like "362*2789" or "4^35".)

Your answer is correct.

Quiz DC3Q

Question 1

Partially correct

Mark 0.60 out of 3.00

[Flag question](#)

Suppose you want to **approximate** the function $f : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = 4e^{3(x+1)} + 4$$

in a **neighborhood of $c = -1$** by a **polynomial function of degree 3**.

How would you choose the **coefficients** in the representation of the polynomial given below?

$$\boxed{4} + \boxed{12}(x+1) + \boxed{36}(x+1)^2 + \boxed{108}(x+1)^3$$

Note, that the pattern uses "+" between all terms. If a "-" **is required** you must write a **negative coefficient**! Enter **decimal numbers** and round to **2 decimal places**, **fractions** like "3/4" can also be used.

Use this polynomial to compute an **approximation** of $e^{1.2}$.

$$e^{1.2} \approx \boxed{0.2/3}$$



The correct answer is: 8, 12, 18, 18, 3.208

Your answer is partially correct.

You have correctly answered 0 part(s) of this question.

Question 2

Incorrect

Mark 0.00 out of 2.00

[Flag question](#)

Given the function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ defined by

$$f(x) = 9 \ln(5x) + 4.$$

Apparently, the derivatives of f are as follows:

$$\begin{aligned} f'(x) &= \frac{9}{x} \\ f''(x) &= -\frac{9}{x^2} \\ f^{(3)}(x) &= \frac{18}{x^3} \\ &\vdots \\ f^{(n)}(x) &= (-1)^{n-1} \frac{9(n-1)!}{x^n} \end{aligned}$$

The Taylor polynomial $T_{11}(x)$ of f at 1 of order 11 is a sum of powers of the form $t_k \cdot (x-1)^k$. What is the coefficient t_7 at power 7 in this Taylor polynomial?

$$t_7 = \boxed{9 \cdot 720}$$



The correct answer is: 1.2857142857143

Your answer is incorrect.

Question 3

Incorrect

Mark 0.00 out of 2.00

[Flag question](#)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function $f(x) = 6x^3 + 5x^2 + 8x + 8$. There is a number B such that f is concave in $(-\infty, B)$ and convex in (B, ∞) .

Compute B .

$$\boxed{36} \times$$

The correct answer is: -0.27777777777778

Your answer is incorrect.

Question 4

Incorrect

Mark 0.00 out of 3.00

 Flag question

We want to use **L'Hopital's rule (LH)** to study the following limit

$$\lim_{x \rightarrow 9} \frac{11 \sin(\pi x)}{x^2 - 81}.$$

Examining this limit, we find

- ☐ LH is not applicable for this problem.
- ☐ LH is applicable for this problem, but not decisive.
- ☐ LH is applicable for this problem, but using LH, we find the limit does not exist.
- ☐ LH is applicable, and using LH, we find the limit is -1 .
- ☐ LH is applicable, and using LH, we find the limit is $\frac{\pi}{2}$.
- ☐ LH is applicable, and using LH, we find the limit is 1 .
- ☒ LH is applicable, and using LH, we find the limit is 0 .
- ☐ LH is applicable, and using LH, we find the limit is $\frac{11}{18}$.
- ☐ LH is applicable, and using LH, we find the limit is $-\frac{11}{18}$.
- ☐ LH is applicable, and using LH, we find the limit is $-\frac{11\pi}{18}$.

✖

The correct answer is: LH is applicable, and using LH, we find the limit is $-\frac{11\pi}{18}$.

Your answer is incorrect.

Question 1

Quiz DC3B

Correct

Mark 1.00 out of 1.00

We want to use **L'Hopital's rule (LH)** to study

$$\lim_{x \rightarrow 2} \frac{-114x^2 + 209x + 19x^3 - 114}{-66 + 11x^3 - 66x^2 + 121x}.$$

Examining this limit, we find

- ☐ LH is not applicable for this problem.
- ☐ LH is applicable for this problem, but not decisive.
- ☐ LH is applicable for this problem, but using LH, we find the limit does not exist.
- ☐ LH is applicable, and using LH, we find the limit is $\frac{11}{19}$.
- ☐ LH is applicable, and using LH, we find the limit is $\frac{-114}{-66}$.
- ☐ LH is applicable, and using LH, we find the limit is $\frac{-66}{-114}$.
- ☐ LH is applicable, and using LH, we find the limit is $-\frac{19}{11}$.
- ☒ LH is applicable, and using LH, we find the limit is $\frac{19}{11}$.
- ☐ LH is applicable, and using LH, we find the limit is 0.



The correct answer is: LH is applicable, and using LH, we find the limit is $\frac{19}{11}$.

Your answer is correct.

Question 2

Correct

Mark 1.00 out of 1.00

Suppose from an "unknown" function f you know

$$f(4) = -5$$

$$f'(4) = -7$$

$$f''(4) = -8$$

$$f^{(3)}(4) = 6$$

$$f(5.81) = -24.84$$

a) If you want to **approximate** $f(x)$ for x near 4 using the given information, you would use a **Taylor polynomial** at of order .

✓

The correct answer is: 4, 3

b) Compute an **approximation** of $f(4.07)$ using the "**best possible**" **Taylor polynomial** (considering the given information).

$$f(4.07) \approx \text{$$

Give the value as a decimal number or provide the formula to compute the value from the given numbers.

✓

The correct answer is: -5.51

Your answer is correct.

Quiz BIT1Q

Question 1

Correct

Mark 2.00 out of 2.00

Flag question

Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = (x^5 + 1)e^{x^6+6x+6}.$$

Consider the function $F(x)$ such that

$$F'(x) = f(x).$$

Which of the following is a possible form of $F(x)$?

- ☐ $(x^5 + 1)^2 e^{x^6+6x+6}$
- ☐ e^{x^6+6x+6}
- ☒ $\frac{1}{6} e^{x^6+6x+6}$ ✓
- ☐ $e^{x^6+6x+6} + \int (x^5 + 1)e^{x^6+6x+3} dx$

Your answer is correct.

The correct answer is: $\frac{1}{6} e^{x^6+6x+6}$

Question 2

Correct

Mark 2.00 out of 2.00

Flag question

Use **Integration by Substitution** to compute

$$I = \int \frac{(\ln(\cos x))^2 \sin x}{\cos x} dx.$$

Which of the following is a correct expression for I ?

- ☒ $-\frac{1}{3} (\ln(\cos x))^3$ ✓
- ☐ $\frac{1}{3} (\ln(\cos x))^3$
- ☐ $\frac{1}{2} (\ln(\cos x))^2$
- ☐ $-(\ln(\cos x))^2$

Your answer is correct.

The correct answer is: $-\frac{1}{3} (\ln(\cos x))^3$

Question 3

Partially correct

Mark 2.40 out of 3.00

Flag question

Consider the **polynomial function** $f : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = \sum_{i=0}^3 5a_i x^i,$$

where the coefficients a_i are as follows:

a_0	a_1	a_2	a_3
5	2	-6	1

Find an **antiderivative** $g = \int f(x) dx$ that satisfies $g(0) = 7$ in the form of a **polynomial function of degree 4**, i.e., a function $g : \mathbb{R} \rightarrow \mathbb{R}$ with

$$g(x) = \sum_{i=0}^4 b_i x^i.$$

Fill in the coefficients of g in the table below.

b_0	b_1	b_2	b_3	b_4
1	25	5	-10	5/4

You can use **fractions** or decimal numbers **rounded to 2 digits**.



The correct answer is: 7, 25, 5, -10, 1.25

Your answer is partially correct.

You have correctly answered 0 part(s) of this question.

Question 4

Partially correct

Mark 1.50 out of 3.00

 Flag questionUse **Integration by Parts** to evaluate the integral:

$$I = \int 3(x+1)^2 \ln(x+1) dx.$$

The result can be expressed in the form:

$$I = A(x+1)^B \ln(x+1) - C(x+1)^D$$

for certain constants **A**, **B**, **C**, and **D**. Calculate the values of **A**, **B**, **C**, and **D**, that complete the expression for the integral **I** and fill them in **all** the fields below.A = B = C = D = 

The correct answer is: 1, 3, 0.33, 3

Note: You may use elementary computation in the input fields e.g. $2*2$ will yield 4 or $3*(2+2)/2$ will yield 6. If you decide to write your answers as decimals, please round them to **two decimal places** (e.g., write 3.23, not 3.2334, and 1.24 not 1.2354).

Your answer is partially correct.

You have correctly answered 0 part(s) of this question.

Question 1

Quiz BIT1B

Partially correct

Mark 0.67 out of 1.00

We calculate $\int x^2(x+10)^{55}dx$ and obtain

$$x^a \frac{(x+10)^{56}}{b} - x^c \frac{(x+10)^{57}}{d} + \frac{(x+10)^e}{f}$$

where

a =

b =

c =

d =

e =

f =

Note: You may use elementary computation in the input fields e.g. $2*2$ will yield 4 or $3*(2+2)/2$ will yield 6.



The correct answer is: 2, 56, 1, 1596, 58, 92568

Die Antwort ist teilweise richtig.

You have correctly answered 0 part(s) of this question.

Question 2

Partially correct

Mark 0.33 out of 1.00

Use **Integration by Parts** to compute

$$I = \int -9x^2 \sin(-7x) dx.$$

The result can be expressed in the form

$$Ax^2 \cos(Bx) + C \cos(Dx) + Ex \sin(Fx)$$

for certain constants **A**, **B**, **C**, **D**, **E**, and **F**, where **B**, **D**, and **F** have **the same sign** as -7.

Calculate the values of **A**, **B**, **C**, **D**, **E**, and **F** and fill them in **all** the fields below.

$$\boxed{-1.29} x^2 \cos(\boxed{7} x) + \boxed{0.05} \cos(\boxed{7} x) + \boxed{0.37} x \sin(\boxed{7} x)$$



The correct answer is: -1.29, -7, -0.37, -7, 0.05, -7

Note: Please write your answers as decimals rounded to **two decimal places** (e.g., write 3.23, not 3.2). Be **careful with signs**, the given formula pattern has "+" everywhere, so you need to put a "-" into the answer field in case of negative coefficients.

Your answer is partially correct.

You have correctly answered 0 part(s) of this question.

Quiz BIT2Q

Question 1

Incorrect

Mark 0.00 out of 3.00

🚩 Flag question

Let $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt{1 + 2^x(63 - 63x)}$.

Compute $\int_0^1 f'(x) dx$.

✖

The correct answer is: -7

Your answer is incorrect.

Question 2

Incorrect

Mark 0.00 out of 3.00

🚩 Flag question

Consider the function $f : [0, 2] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 14x & 0 \leq x \leq 1 \\ 28x - 14 & 1 \leq x \leq 2 \end{cases}$$

Compute the lower sum $L_2(f)$

 ✖

The correct answer is: 14

Your answer is incorrect.

Question 3

Incorrect

Mark 0.00 out of 2.00

🚩 Flag question

If the graph of the linear function $f : [2, 4] \rightarrow \mathbb{R}, f(x) = 3x$ rotates around the x-axis, it generates a truncated cone.

The formula for the volume of a body of revolution generated by a rotating graph of f is $\pi \int_a^b (f(x))^2 dx$. Compute the volume of the truncated cone divided by π .

 ✖

The correct answer is: 168

Your answer is incorrect.

Question 4

Incorrect

Mark 0.00 out of 2.00

🚩 Flag question

Calculate the following integral:

$$I = \int_1^{16} \frac{4}{\sqrt{x}} dx.$$

✖

The correct answer is: 24

Your answer is incorrect.

Question 1

Quiz BIT2B

Correct

Mark 1.00 out of 1.00

Calculate the following integral:

$$I = \int_2^{19.5} \frac{2}{\sqrt{2x-3}} dx.$$



The correct answer is: 10

Your answer is correct.

Question 2

Correct

Mark 1.00 out of 1.00

Let $f : [0, 1] \rightarrow \mathbb{R}$ be the function $f(x) = 5x^2 + 3x + 4$.Compute the upper sum $U_4(f)$.

The correct answer is: 8.21875

Your answer is correct.

Quiz BIT3Q

Question 1

Incorrect

Mark 0.00 out of 3.00

[Flag question](#)

A random quantity has probability density function $p : \mathbb{R} \rightarrow \mathbb{R}, p(x) = e^{-|x|}/2$.

Compute $\mathbb{P}(x \in [-\ln(2), \ln(2)])$.

✖

The correct answer is: 0.5

Your answer is incorrect.

Question 2

Correct

Mark 3.00 out of 3.00

[Flag question](#)

Compute $\int_3^4 (x-3)^{-0.6} dx$.

✔

The correct answer is: 2.5

Your answer is correct.

Question 3

Incorrect

Mark 0.00 out of 4.00

[Flag question](#)

The graph of the function $f : [e^4, e^{12}] \rightarrow \mathbb{R}, f(x) = 5/\sqrt{x}$ rotates around the x-axis.

The volume of revolution is given by

$$V = \pi A$$

Determine the rational number A.

A=

✖

The correct answer is: 200

Your answer is incorrect.

Question 1

Quiz BIT3B

Correct

Mark 1.00 out of 1.00

A random quantity X is uniformly distributed in $[3, 8]$. Compute the expectation value of $(X - 1)^2$.

 ✓

The correct answer is: 22.333333333333

Your answer is correct.

Question 2

Incorrect

Mark 0.00 out of 1.00

Compute $\int_1^\infty \frac{1}{(x+2)^{2.3}} dx$.

 ✗

The correct answer is: 0.18441617777561

Your answer is incorrect.

Quiz MC1Q

Question 1

Correct

Mark 3.00 out of 3.00

Flag question

Consider the vector

$$v = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}.$$

Fill in the appropriate norm of vectors in the table below.

$\ v\ _2$	$\ v\ _1$	$\ v\ _\infty$
3	5	4

You can use **fractions** or decimal numbers **rounded to 2 digits**.



The correct answer is: 3, 5, 4

Your answer is correct.

Question 2

Correct

Mark 3.00 out of 3.00

Flag question

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = \begin{cases} \frac{2x+y+7}{x^2+2y^2+1}, & \text{if } (x, y) \neq (-2, -1), \\ c, & \text{if } (x, y) = (-2, -1), \end{cases}$$

for a real parameter c . Determine c such that the function f is continuous everywhere.

You can use **fractions** or decimal numbers **rounded to 2 digits**.

$c =$



The correct answer is: 0.28571428571429

Your answer is correct.

Question 3

Correct

Mark 4.00 out of 4.00

Flag question

Determine the limit

$\lim_{k \rightarrow \infty} \mathbf{s}^{(k)}$, where $\mathbf{s} : \mathbb{N} \rightarrow \mathbb{R}^4$ and

$$\mathbf{s}^{(k)} = \left((-2) \cdot \sqrt[4]{2}, \frac{3k^2 - 4k + 4}{2k^2 + 5k + 8}, \sin\left(\frac{8}{k}\right), \cos\left(\frac{4}{k}\right) \right).$$

Denote the limit by $\mathbf{y} = (y_1, y_2, y_3, y_4)$.

Fill in the appropriate numbers in the table below.

y_1	y_2	y_3	y_4
-2	6/4	0	1

You can use **fractions** or decimal numbers **rounded to 2 digits**.



The correct answer is: -2, 1.5, 0, 1

Your answer is correct.

Question 1

Quiz MC1B

Correct

Mark 1.00 out of 1.00

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x_1, x_2) = \begin{cases} \frac{3(x_1^2 - x_2^2)}{x_1^2 + x_2^2}, & \text{if } \mathbf{x} \neq (0, 0), \\ 0, & \text{if } \mathbf{x} = (0, 0). \end{cases}$$

Furthermore, let $s : \mathbb{N} \rightarrow \mathbb{R}^2$ be defined as

$$s^{(k)} = \left(k^{-1} \cdot \cos\left(\frac{1}{2} \cdot \pi\right), k^{-1} \cdot \sin\left(\frac{1}{2} \cdot \pi\right) \right).$$

Determine the following limit.

You can use **fractions** or decimal numbers **rounded to 2 digits**.

$$\lim_{k \rightarrow \infty} f(s^{(k)}) = \boxed{-3}$$



The correct answer is: -3

Your answer is correct.

Question 2

Partially correct

Mark 0.67 out of 1.00

Determine the limit

 $\lim_{k \rightarrow \infty} \mathbf{s}^{(k)}$, where $\mathbf{s} : \mathbb{N} \rightarrow \mathbb{R}^3$ and

$$\mathbf{s}^{(k)} = \left((-2) \cdot \sqrt[k]{6 \cdot k}, \frac{3k^2 - 4k + 4}{(4k - 5)(k + 8)}, \cos\left(\frac{4}{k}\right) + \ln\left(\left(\frac{k+1}{k}\right)^{6k}\right) \right).$$

Denote the limit by $\mathbf{y} = (y_1, y_2, y_3)$.

Fill in the appropriate numbers in the table below.

y_1	y_2	y_3
0	0.75	7

You can use **fractions** or decimal numbers **rounded to 2 digits**.

The correct answer is: -2, 0.75, 7

Your answer is partially correct.

You have correctly answered 0 part(s) of this question.

Quiz MC2Q

Question 1

Incorrect

Mark 0.00 out of 4.00

[Flag question](#)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = 2e^{xy} + 3x \sin(y - 3)$.

Determine the tangent plane of the function f at the point $\mathbf{P}=(0,3)$.

$$T(x, y) = 6 \text{ } + 3 \text{ } x + 3 \text{ } (y - 3)$$

✖

The correct answer is: 2, 6, 0

Your answer is incorrect.

Question 2

Incorrect

Mark 0.00 out of 4.00

[Flag question](#)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = 26 \cdot \sqrt{x^2 + 3y^2}.$$

We wish to determine the directional derivative $D_{\mathbf{u}}f(2, 2)$ when the direction is given by $\mathbf{u} = (\frac{12}{13}, \frac{5}{13})$.

1. We wish to use the *gradient formula* for differentiable functions and therefore we need first the 1st order partial derivatives. Compute them.

(You can use **fractions** or decimal numbers **rounded to 2 digits**.)

$$\frac{\partial f}{\partial x}(2, 2) = 12$$

✖

The correct answer is: 13

$$\frac{\partial f}{\partial y}(2, 2) = 78$$

✖

The correct answer is: 39

2. Finally, by the *gradient formula*, compute the directional derivative.

(You can use **fractions** or decimal numbers **rounded to 2 digits**.)

$$D_{\mathbf{u}}f(2, 2) = (12 \cdot (12/13)) + ($$

✖

The correct answer is: 27

Your answer is incorrect.

Question 3

Correct

Mark 2.00 out of 2.00

[Flag question](#)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = 2x^2y^2 + 7y^2$.

Compute all the second-order partial derivatives of f , that is:

$$\frac{\partial^2 f}{\partial x \partial y}(x, y) = 8 \text{ } xy$$

$$\frac{\partial^2 f}{\partial y \partial x}(x, y) = 8 \text{ } xy$$

$$\frac{\partial^2 f}{\partial x \partial x}(x, y) = 4 \text{ } y^2$$

$$\frac{\partial^2 f}{\partial y \partial y}(x, y) = 4 \text{ } x^2 + 14$$

✓

The correct answer is: 8, 8, 4, 4, 14

Your answer is correct.

Question 1

Correct

Mark 1.00 out of 1.00

Quiz MC2B

Compute the Hessian of the following function at the point $\mathbf{P}_0 = (1, 0)$.

$$f(x, y) = \cos(3y) - 3x^2y - 5x^3.$$

For entering the Hessian, we define $H_f(\mathbf{P}_0) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$a =$

$b =$

$c =$

$d =$



The correct answer is: -30, -6, -6, -9

(You can use **fractions** or decimal numbers **rounded to 2 digits**.)

Your answer is correct.

Question 2

Correct

Mark 1.00 out of 1.00

Determine if the vector $\left(\frac{2}{5}, \frac{2}{5}\right)$ define a direction in $\mathbb{R}^2$.

☐ yes☒ no

Determine if the vector $\left(\frac{3}{5}, \frac{4}{5}\right)$ define a direction in $\mathbb{R}^2$.

☒ yes☐ no

Determine if the vector $\left(-\frac{5}{4}, \frac{1}{2}, \frac{1}{2}\right)$ define a direction in $\mathbb{R}^3$.

☐ yes☒ no

Determine if the vector $\left(\frac{4}{9}, \frac{4}{9}, \frac{7}{9}\right)$ define a direction in $\mathbb{R}^3$.

☒ yes☐ no

The correct answer is: no, yes, no, yes

Your answer is correct.